

SATQUERY AI — REMOTE SENSING ANALYSIS REPORT

ISRO Problem Statement 26167 • Mission ID: a61f3b65 • Generated: 17:54:45.863Z

Natural Query:	Identify and compute the area of the launch complexes and propellant facilities.
Classified Task:	GROUNDING
Calibrated Confidence:	90% (Spatial Intersection & IoU Match)
Specialist Models:	satquery-grounding-rs-v1
Total Latency:	153.4 ms

GROUNDING AI INTERPRETATION

Successfully grounded 'launch complex / launch pad'. Identified 3 prominent region(s) spanning a total surface area of 69.58 hectares (0.6958 km²), centered at approximately 13.733643° N, 80.225635° E (3.92% of the satellite footprint).
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INPUT SENSOR METADATA

Filename	Modality	Dimensions	Bands	GeoTIFF CRS
isro_sdsc_spaceport.	OPTICAL	512x512	3	EPSG:4326

DELINEATED VECTOR TARGETS & SPATIAL TELEMETRY

Feature	Area (ha)	Area (km²)	Perimeter (m)	Centroid [Lat, Lon]	Confidence
Launch complex / launc	35.60	0.3560	2396.5	13.733643°, 80.225635°	80%
Launch complex / launc	16.99	0.1699	3723.1	13.708154°, 80.243408°	80%
Launch complex / launc	16.99	0.1699	3723.1	13.741943°, 80.243408°	80%

OBSERVABLE AGENTIC EXECUTION TRACE

#	Phase / Event	Observable Details	Time (m)
1	Query Ingestion	Received natural language query: 'Identify and compute the area of the launch complexes and propellant facilities.' with 1 uploaded image(s).	0.0
2	Input Validation	Verified 1 image file(s). Format and size limits satisfied.	0.0
3	Raster Metadata Extraction: isro_sdsc_spaceport.tif	Extracted dimensions (512x512), 3 band(s). CRS: EPSG:4326	59.0
4	Modality Identification	Detected sensor modalities: ['OPTICAL'].	0.0
5	Query Classification → GROUNDING	Single image detected with spatial referring/localization/grounding intent -> Routed to Text-Guided Grounding Specialist.	0.0

6	Model Dispatch	Dispatched task to specialist: 'satquery-grounding-rs-v1' on device 'cpu'.	0.0
7	Specialist Execution Completed	Inference finalized in 94.0ms. Evidence artifacts compiled.	94.0
8	Result Fusion & Calibration	Derived grounded response with calibrated confidence 90% (Spatial Intersection & IoU Match).	0.0